

My article submission is included starting on the next page. The first 19 pages have a journal-style layout, which I believe is more readable. This is followed by identical content using double-spacing.

The $\text{\LaTeX}$ sources for this paper were generated from my own format, which uses syntax vaguely similar to Markdown. Apart from my finding it easier to compose documents in this format — which I have used in the past for published books and articles — it has the additional feature of directly producing JATS files. I have found that some Diamond Open-Access journals openly publish JATS alongside PDF files, but this practice is not adopted consistently. As a fallback, authors could provide JATS representations in “Supplemental Materials” or other data available in conjunction with a publication.

To that end, I have been working on a code library for manipulating JATS files. For at least my own use, this library includes extensions to JATS to address issues such as improved PDF interop (noting pdfpage coordinates of textual landmarks, for example) and data set cross-references — i.e., tags for describing data set content, similar to Research Object specifications, as well as annotating text segments relevant to specific parts/aspects of the data set (statistical parameters, units of measurement, algorithms, formulae/equations, table columns, etc.).

Another feature is sentence-level tags to enable bibliographic queries such as merging queries on matches that appear in the same sentence. Via PDF integration, GUI tools could provide functionality such as highlighting whole sentences in conjunction with query results, or in response to user gestures. In general, directly representing sentences and sentence-boundaries is conducive to text-mining and other computational treatment of published text. Similarly, these JATS extensions allow authors to fine-tune word-vectors, token-streams, and stemming/lemmatization policies for automatic indexing. It seems that full-text search engines do not ingest JATS sources directly (apparently relying on JATS-to-HTML pipelines instead, which lose important context), which is another reason for a JATS processing library (I hope to patch it into search engine frameworks, like Pisa). New JATS tags are also appropriate for manual indexing, to support more granular and semantically modeled associations between index terms and their locators.

I am particularly interested in JATS extensions for linguistics, because this discipline has numerous kinds of domain-specific textual content, such as example sentences subject to analysis. Working in linguistics papers in the past, I have noticed a lack of formal data models for issues such as declaring different aspects of such sentence-examples, including notation for gradations of acceptability, contextual indications (often set apart via square brackets), and alternatives for words or phrases. Conventions seem fairly well-established for text itself, but not for how these various details should be represented in machine-readable sources like XML or $\text{\LaTeX}$.

With proper encoding, it would be possible to construct “corpora” of sample sentences (or other units on different scales) analyzed in linguistic publications. Such corpora could be searchable, and trace where a particular example is addressed by multiple authors. With a dedicated PDF viewer, query results related to specific numbered examples could show the proper page while also highlighting and/or outlining the specific example. More advanced analytic diagrams (e.g., Link Grammar parsing summaries) could be represented in separate windows, opened in response to context menus activated proximate to corresponding sentence-examples.